DANIEL MORA

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SUMMARY

Computer Science and AI graduate with expertise in machine learning, LLMs and quantitative research. Experienced in building and shipping ML systems end to end, from point-in-time data platforms and custom GPU kernels to autonomous agentic pipelines, with released models and a sole-author paper. Strong background in deep learning, NLP and data-driven decision-making from hedge-fund research.

EDUCATION

IE University — **Madrid, Spain**
Bachelor in Computer Science and Artificial Intelligence — 2026

EXPERIENCE

WhiteHole | *Hedge fund* — **San Sebastián, Spain**
Quantitative Research Intern — Jun 2025 – Nov 2025

- Built CPI and GDP forecasts from housing-market indicators with **strong out-of-sample accuracy**.
- Showed that NLP signals from political communications **predict market volatility** across tree, linear and ridge models, and **identified a regime shift** as markets grew less sensitive to them.
- Built a knowledge graph of company behavior with NLP extraction and local LLMs; **used as a trading signal** and **adopted by the private-equity team** through an MCP server.

MAI-Book | *E-commerce start-up* — **Remote**
Front-end Developer Intern — Jun 2023 – Oct 2023

- **Cut client churn by about 20%** by analyzing user-testing data and refining the online library's sales funnel.
- Built user-facing web interfaces and ran A/B tests and UX research to guide product and conversion changes.

RESEARCH AND PROJECTS

JEPA-Embedding Transformer (ArgusJEPA + CassaFormer) | *PyTorch, Triton* — 2026

- Self-supervised JEPA embeddings feed a 2.05M-parameter MoE Transformer; **code and weights released**.
- Wrote a custom Triton MoE kernel: **up to 1.5× end-to-end speedup**.
- Built a point-in-time, survivorship-free data platform for the top 500 US stocks (prices, filings, macro, news).

TC_ACC: Agentic Documentary Studio | *Python, LangGraph, Pydantic* — 2026

- Turns one link into a **finished 10-minute documentary in about five hours, with no human in the loop**.
- Episodes have drawn **400+ watch hours on YouTube**; 28 stages across 5 departments, with 2 review gates.

Brain–LLM Sparse Feature Correspondence (BA Thesis) | *PyTorch, SAEs, fMRI* — 2026

- Showed that SAE features **predict held-out brain responses** (59 participants, six SAEs) but that brain and model **do not rely on them in the same order** ($\rho = -0.06$); written up as a **sole-author paper**.

Sparse Autoencoder Releases | *PyTorch, Hugging Face, GCP*

- OLMo-3 SAEs: **released seven TopK sparse autoencoders** across four layers of OLMo-3-7B-Think.
- Triplet extractor: SAE steering vs. LoRA; the fine-tuned Gemma 3 4B gets **about 50 monthly downloads**.

LEADERSHIP

- **Founder and President**, IE Dominican Republic Club: built IE's largest student-led diversity event (500+).
- **President**, IE Quant Finance Club.
- **Business Development Officer**, IE Investment Club.

SKILLS

ML:	PyTorch, Triton, Hugging Face, JEPA, sparse autoencoders, state-of-the-art LLM architectures
Modeling:	Time-series forecasting, linear/ridge/tree models, clustering, Granger causality, regime analysis
Agents:	LangGraph, Pydantic, multi-stage agent workflows, RAG, GraphRAG, local LLM inference
Systems:	Python, C++, SQL, PostgreSQL, DuckDB, Redis, FastAPI, Docker, GitHub Actions, GCP, Azure
Languages:	Spanish (native), English (fluent)